ShARC: Section 3 — Problem Formulation

Risk-Sensitive RL for Hierarchical Arc Routing under Compound Distribution Shifts

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1 Problem Formulation

We develop a formal model for the *Hierarchical Capacitated Arc Routing Problem under Distribution Shifts* (HCARP-DS). We first define the base combinatorial problem (§1.1), then introduce the distribution shift model (§1.2), and finally cast HCARP-DS as a Markov Decision Process suitable for risk-sensitive reinforcement learning (§1.3).

1.1 The Hierarchical Capacitated Arc Routing Problem

Definition 1 (HCARP Instance)**.** *An HCARP instance is a tuple* $\langle G, K, Q, \ell, c, d\rangle$ *where:*

- $G = (V, E)$ *is a weighted undirected graph with vertex set* V *(including a depot* $v_0 \in V$*) and edge set* $E = E_r \cup E_d$*, where* $E_r \subseteq E$ *is the set of* required arcs *and* $E_d = E \setminus E_r$ *the set of* dead-head arcs *used only for traversal.*
- $K \in \mathbb{N}$ *is the fleet size (number of homogeneous vehicles).*
- $Q > 0$ *is the vehicle capacity (shared by all vehicles).*
- $\ell : E_r \to \{1, 2, 3\}$ *assigns each required arc a* priority class. *Class 1 arcs are highest priority.*
- $c : E \to \mathbb{R}_{>0}$ *is the traversal cost function (time or distance).*
- $d : E_r \to \mathbb{R}_{>0}$ *is the service demand function.*

A *solution* to an HCARP instance is a set of K routes $\Pi = \{\pi_1, \ldots, \pi_K\}$, where each route π_k is a closed walk starting and ending at v_0. The routes must collectively cover every required arc in E_r exactly once. Each route must respect the capacity constraint: $\sum_{e \in \pi_k \cap E_r} d_e \leq Q$.

Lexicographic objective. For priority class $j \in \{1, 2, 3\}$, define the *class completion time* $T_j(\Pi)$ as the makespan of the last vehicle to finish all class-j arcs:

$$T_j(\Pi) \;=\; \max_{k=1}^{K} \sum_{\substack{e \in \pi_k \\ \ell(e) \le j}} c_e \,. \tag{1}$$

The primary objective is to minimise $T_{\max}(\Pi) = \max(T_1, T_2, T_3)$, which penalises the worst-delayed priority class. In case of ties on $T_{\max}$, we break ties by minimising T_1, then T_2, then T_3. Formally, the objective is lexicographic minimisation of the vector (T_1, T_2, T_3) ordered by max-priority:

$$\min_{\Pi} \big(T_{\max}(\Pi),\; T_1(\Pi),\; T_2(\Pi),\; T_3(\Pi)\big). \tag{2}$$

Remark 1. *The lexicographic structure makes HCARP strictly harder than standard CARP, since the priority ordering couples the routing decisions across all classes. In particular, a route assignment that minimises total cost may not minimise $T_{\max}$, and vice versa. The non-smooth max operator in the objective also complicates direct policy gradient estimation; we address this in §1.3.*

1.2 Distribution Shift Model

Real-world arc routing operates under non-stationary conditions: road service demands vary by season, traversal costs fluctuate with traffic patterns, and task availability changes due to cancellations or priority re-assignments. We capture this jointly as a *compound distribution shift* parameterised by a continuous shift vector.

Definition 2 (Shift Vector)**.** *A shift vector $\boldsymbol{\phi} = (\delta_d, \delta_c, \delta_p) \in [0, 1]^3$ parameterises three simultaneous axes of distributional drift:*

- $\delta_d \in [0, 1]$*: demand drift. The service demand of each required arc is scaled by a factor drawn from $d_e^{(\boldsymbol{\phi})} \sim d_e \cdot \mathrm{Uniform}(1, 1 + \delta_d)$.*
- $\delta_c \in [0, 1]$*: cost drift. Traversal costs are perturbed as $c_e^{(\boldsymbol{\phi})} \sim c_e \cdot \mathrm{Uniform}(1, 1 + \delta_c)$.*
- $\delta_p \in [0, 1]$*: availability drift. Each required arc $e \in E_r$ is independently available with probability $1 - \delta_p$; with probability δ_p the arc is removed from E_r for that episode.*

Definition 3 (HCARP-DS Instance)**.** *An HCARP-DS instance is an HCARP instance $\langle G, K, Q, \ell, c, d\rangle$ equipped with a shift distribution $\subseteq [0, 1]^3$ from which a shift vector $\boldsymbol{\phi}^{(i)}$ is drawn at the start of each episode i:*

$$\boldsymbol{\phi}^{(i)} \;\sim\; \mu_\Phi \;\; \textit{with support} \;\; . \tag{3}$$

The realised instance for episode i is $\langle G, K, Q, \ell, c^{(\boldsymbol{\phi}^{(i)})}, d^{(\boldsymbol{\phi}^{(i)})}, E_r^{(\boldsymbol{\phi}^{(i)})}\rangle$, where the superscript denotes application of the shift.

Non-rectangularity of the uncertainty set. A central structural property of the HCARP-DS shift model is that $\subseteq [0, 1]^3$ is *non-rectangular*: the joint perturbation $(\delta_d, \delta_c, \delta_p)$ cannot in general be decomposed into independent per-arc, per-dimension perturbations. Specifically, the availability shift δ_p acts on the set E_r globally and affects the feasibility structure of all routes simultaneously. The demand and cost shifts δ_d, δ_c act multiplicatively on arc-level attributes but are correlated through the shared shift draw $\boldsymbol{\phi}^{(i)}$.

Remark 2 (Contrast with Rectangular UCARP)**.** *The classical Uncertain CARP (UCARP) of Mei et al. [1] models uncertainty as independent stochastic demand realisations $d_e \sim \mathcal{D}_e$ for each arc e, drawn from fixed distributions. This is a* rectangular *uncertainty model: the adversary selects demand realisations independently per arc. The HCARP-DS shift model is non-rectangular because (i) the three shift axes are drawn jointly from μ_Φ, (ii) the availability shift couples arc existence decisions across the entire instance, and (iii) the shift magnitude $\boldsymbol{\phi}$ itself drifts across episodes rather than being fixed. This breaks the standard UCARP dynamic programming decomposition, motivating the MDP formulation in §1.3.*

Shift curriculum. To stabilise training, we anneal the maximum shift magnitude over the course of training. Let $\Delta_{\max} = (\delta_d^{\max}, \delta_c^{\max}, \delta_p^{\max})$ be the target shift budget. During training episode i, the effective shift budget is:

$$\Delta^{(i)} \;=\; \Delta_{\max} \cdot \min\left(\frac{i}{I_{\text{warm}}},\, 1\right), \tag{4}$$

where I_{warm} is a warmup horizon. The policy is first trained on nominally-distributed instances, then progressively exposed to larger shifts as training matures. At evaluation time, $\boldsymbol{\phi}$ is drawn from the full to assess worst-case tail performance.

1.3 HCARP-DS as a Budget-Augmented MDP

We now cast HCARP-DS as an MDP that admits risk-sensitive policy optimisation via CVaR-REINFORCE.

Definition 4 (HCARP-DS MDP)**.** *The HCARP-DS MDP is a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where:*

***State space** $\mathcal{S}$. At decision step t, the state s_t encodes:*

- Graph representation: *arc-level features $(c_e^{(\boldsymbol{\phi})}, d_e^{(\boldsymbol{\phi})}, \ell_e, \mathbf{1}[e \in E_r^{(\boldsymbol{\phi})}])$ for all $e \in E$, computed under the current episode's shift $\boldsymbol{\phi}$.*
- Vehicle state: *for each vehicle k, its current position $v_k \in V$, remaining capacity $q_k \leq Q$, and elapsed time t_k.*
- Coverage indicator: *a binary vector $m \in \{0,1\}^{|E_r|}$ indicating which required arcs have been serviced.*
- Budget variable: $\beta_t \in [0,1]$, *a running estimate of the empirical CVaR quantile used to make the CVaR objective tractable (defined in §1.3).*

***Action space** $\mathcal{A}$. At each step, the active vehicle selects one of:*

$$a_t \;\in\; \{0\} \cup \{1, \ldots, |E_r|\}, \tag{5}$$

where $a_t = 0$ signals a vehicle switch (the current vehicle returns to depot and the next vehicle is dispatched), and $a_t = e$ signals that arc e is serviced next. Infeasible actions (capacity-violating or already-serviced arcs) are masked out by the policy.

***Transition** $\mathcal{P}$. Transitions are deterministic given the shift realisation $\boldsymbol{\phi}^{(i)}$ for that episode. Taking action a_t updates the vehicle position, remaining capacity, elapsed time, and coverage indicator.*

Reward $\mathcal{R}$. *We use a* terminal reward*:*

$$r_t \;=\; \begin{cases} -T_{\max}(\Pi) & \textit{if } t = T \textit{ (episode end)}, \\ 0 & \textit{otherwise.} \end{cases} \tag{6}$$

Using terminal reward avoids reward shaping artefacts and ensures the policy optimises the true lexicographic objective. The discount factor is $\gamma = 1$ *(episodic, undiscounted).*

CVaR objective. Let $Z^\pi(\boldsymbol{\phi}) = T_{\max}(\Pi^\pi; \boldsymbol{\phi})$ denote the episode cost (makespan) under policy π and shift $\boldsymbol{\phi}$. Standard risk-neutral RL minimises $\mathbb{E}_{\boldsymbol{\phi}[Z^\pi(\boldsymbol{\phi})]}$, which is sensitive to distributional shifts in the tail. We instead optimise the *Conditional Value-at-Risk at level* α:

$$\mathrm{CVaR}_\alpha(Z^\pi) \;=\; \min_{\beta \in \mathbb{R}} \left\{ \beta + \frac{1}{1-\alpha}\, \mathbb{E}_{\boldsymbol{\phi}[\max(Z^\pi(\boldsymbol{\phi}) - \beta,\, 0)]} \right\}, \tag{7}$$

where $\alpha \in (0, 1)$ is the risk level and $\beta^* = \mathrm{VaR}_\alpha(Z^\pi)$ is the Value-at-Risk. CVaR_α measures the expected cost in the worst $(1 - \alpha)$ fraction of episodes, making it sensitive to tail failures under severe distribution shifts.

The budget variable β_t augmented into the state is a running sample estimate of $\mathrm{VaR}_\alpha(Z^\pi)$ maintained across a mini-batch of episodes. This follows the approach of [3], which shows that the inner minimisation over β in (7) can be approximated via a sample average when β is included as a learnable or estimated quantity.

CVaR-REINFORCE gradient estimator. Let $\tau = (s_0, a_0, \ldots, s_T)$ denote a trajectory under policy π_θ and let $R(\tau) = -Z^\pi(\boldsymbol{\phi})$ be the negative episode cost. The gradient of the CVaR objective with respect to θ is:

$$\nabla_\theta \,\mathrm{CVaR}_\alpha(Z^\pi) \;=\; -\,\mathbb{E}_\tau[\mathbf{1}[R(\tau) \le -\beta] \cdot \nabla_\theta \log \pi_\theta(\tau) \cdot R(\tau)] \cdot \frac{1}{1-\alpha}, \tag{8}$$

where $\mathbf{1}[\cdot]$ is the indicator function that selects only the $(1 - \alpha)$ worst trajectories (those with return below $-\beta$). In practice, β is estimated as the empirical α-quantile of returns within each mini-batch. The estimator (8) is the policy gradient analogue of the CVaR gradient derived in [2], adapted to the REINFORCE framework.

Remark 3 (Connection to CMDP relaxation)**.** *An alternative formulation casts the CVaR objective as a Constrained MDP (CMDP): maximise* $\mathbb{E}[R(\tau)]$ *subject to* $\mathrm{CVaR}_\alpha(R(\tau)) \ge -C$ *for a budget* C*. The primal-dual relaxation of this CMDP guarantees constraint satisfaction in expectation but not in the tail. The CVaR-REINFORCE estimator in* (8) *directly optimises the tail objective without the CMDP relaxation, at the cost of higher variance. Whether tail-level constraint satisfaction guarantees can be recovered under non-rectangular compound shifts is an open theoretical question that we discuss in §6.*

References

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